

Introduction to Swaps

BUSS386. Futures and Options

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Lecture Outline

- How a swap works: one contract standing in for a whole stack of forward rate agreements
- Why counting days matters — the conventions that decide what actually changes hands
- Treasury and SOFR futures: the cheapest-to-deliver game, hedging a bond book, and life after LIBOR
- What sets the fixed rate — and why the floating leg is worth par at every reset
- Why the market exists at all, and the swaps built for the awkward cases
- Currency, commodity and total-return swaps: same logic, different underlyings

Why this lecture matters

- OTC derivatives outstanding (BIS): notional ~\$700T globally. The largest single product family inside that figure is **interest-rate swaps**.
- In Lec02 we built a duration-matched dealer hedge with cash bonds. In practice, virtually *all* of that hedging is done with **IRS** — a stack of forward rate agreements bundled into one contract.
- Today: how the contracts work, how they're priced, the close cousins (currency swaps, commodity swaps, total-return swaps), and the post-LIBOR floating leg (SOFR, KOFR).

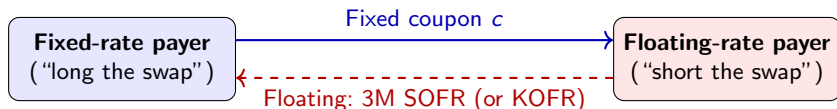
Sources: BIS OTC stats <https://www.bis.org/statistics/derstats.htm>

Swaps

Swap basics

- A swap is a contract calling for an exchange of payments, on one or more future dates, determined by the difference in two reference prices or interest rates.
- A single-payment swap is equivalent to a cash-settled forward contract.
- A swap provides a means to hedge or speculate on a stream of risky cash flows.
- Traded in over-the-counter market.

Plain-vanilla interest-rate swap — the structure



Notional principal N (**not** exchanged)
Net cash flow on each date = $N \cdot (r_{\text{floating}} - c) \cdot \tau$
where τ is the day-count fraction of the period (e.g. ≈ 0.5 for semi-annual)

- Two parties exchange a stream of fixed coupons for a stream of floating coupons, on the same notional, at regular intervals over the swap's life.
- No principal exchange. Only the *net interest* changes hands.
- An IRS is economically a stack of **forward rate agreements (FRAs)** bundled into one contract.

Day Counting

Day-count conventions — why we care

A pre-computer-era inheritance: how to count days for accrued interest. Conventions take the form X/Y (X = days in month, Y = days in year):

Convention	Where it's used	Notes
Actual/Actual	US Treasuries, Australia	"True" day count
30/360	US corp & muni bonds, Eurobonds	360-day year, 30-day month
Actual/360	US money market (incl. SOFR)	Inflates rates slightly
Actual/365	Korea (KTB) , UK, Japan	365-day year

- The catch: inside a *single* swap the fixed and floating legs often use different conventions. We price that mismatch two slides from now.

Reference: <https://www.rbcits.com/en/gmi/global-custody/market-profiles.page>

Day count in practice

4% annual coupon, semi-annual, \$100 face. Partial period March 1 → July 3 (124 actual days; 122 days under 30/360; full half-period: 184 actual / 180 under 30/360):

Convention	Calculation	Accrued (per \$100)
Act/Act	$\frac{124}{184} \times \$2$	\$1.348
30/360	$\frac{122}{180} \times \$2$	\$1.356
Act/365	$\frac{124}{365} \times 4\% \times \100	\$1.359
Act/360	$\frac{124}{360} \times 4\% \times \100	\$1.378

Useful equivalence

A rate quoted on Act/360 is *not* the same as the same rate on Act/365:

$$5\% \text{ Act/360} = 5\% \times \frac{365}{360} \approx 5.069\% \text{ Act/365.}$$

Excel: =DAYS(end, start) for actual; =DAYS360(start, end) for 30/360.

Why we care: day-count basis inside a single IRS

A USD IRS quotes a single fixed rate, but its two legs use *different* day-count conventions. Same 5% rate on \$1M notional, one quarterly period (90 days under 30/360; 92 actual days):

Leg	Quarterly cash flow
Fixed leg, 30/360: $\$1M \times 5\% \times 90/360$	\$12,500
Floating leg, Act/360: $\$1M \times 5\% \times 92/360$	\$12,778
Day-count gap	\$278

- Over a *full year* the gap is $\$1M \times 5\% \times \left(\frac{365}{360} - \frac{360}{360}\right) =$ \$694
- KRW IRS quotes *both* legs on Act/365 (matching KTB convention)

Interest Rate Futures

Treasury futures — specs and quotes

Underlying. A “virtual” bond. The party with the short position can deliver any of a basket of eligible bonds:

- Ultra T-Bond: > 25y T-Bond: 15–25y 10y note: 6.5–10y
- Benchmark coupon: **6%** (US, since 1999); **5%** for KTB futures (Korea)

Quote conventions.

- US (CME): *thirty-seconds* of a dollar per \$100 face. e.g.,
 $134'215 = 134 + 21.5/32 = 134.671875$.
- Korea (KRX): decimals.

KRX KTB futures

KRX lists 3-year, 5-year, 10-year, and 30-year KTB futures. All *cash-settled* against a basket of 3 eligible delivery bonds. Notional ~~₩~~\$100M per contract on the 3-year; ~~₩~~\$50M on the 10-year and 30-year. Heavily traded by domestic banks and pension funds for duration management.

Bond quotes — clean vs. dirty price

- Quoted price (a.k.a. **clean price**) is for \$100 face value. e.g., a quote of 90'05 means $(90 + 5/32) \times 1,000 = \$90,156.25$ on \$100,000 face.
- The **cash** (or **dirty**) price is what actually changes hands:

$$\text{Cash} = \text{Quoted} + \text{Accrued interest.}$$

Example. March 5, 2013: 11% coupon T-bond maturing July 10, 2028, quoted 95'16 = \$95.50.

- Last coupon: Jan 10, 2013. $\Rightarrow$ Next: Jul 10, 2013.
- Days Jan 10 $\rightarrow$ Mar 5: 54. Days Jan 10 $\rightarrow$ Jul 10: 181.
- Semiannual coupon: $\$100 \times 0.11/2 = \5.50 .
- Accrued interest: $5.50 \times 54/181 = \$1.64$.
- Cash price (per \$100 face): $95.50 + 1.64 = \boxed{\$97.14}$. On \$100,000 face: \$97,140.

Treasury futures — settlement and the CTD bond

Invoice formula (cash received by the short upon delivery):

$$\text{Cash} = \text{Settlement price} \times \text{Conversion factor (CF)} + \text{Accrued interest}.$$

Cheapest-to-deliver (CTD). The short can pick which bond to deliver. Choose the one minimizing the cost net of the invoice received:

$$\text{Cost of delivery} = \underbrace{\text{Quoted bond price}}_{\text{cost to acquire}} - \underbrace{\text{Settlement price} \times \text{CF}}_{\text{received from invoice}}.$$

Quick example. Settlement 93.25, three candidates:

Bond	Quoted	CF	Cost = Q – Sett. × CF	
1	99.50	1.0382	$99.50 - 93.25(1.0382) = 2.69$	
2	143.50	1.5188	$143.50 - 93.25(1.5188) = 1.87$	← CTD
3	119.75	1.2615	$119.75 - 93.25(1.2615) = 2.12$	

Conversion factor — definition and intuition

Definition. The CF is the price the deliverable bond would have if priced to yield 6% (the contract's benchmark coupon).

Intuition.

- Coupon $> 6\%$: bond is richer than benchmark $\Rightarrow CF > 1$
- Coupon $= 6\%$: matches benchmark $\Rightarrow CF = 1$.
- Coupon $< 6\%$: bond is leaner $\Rightarrow CF < 1$

Why this design (basket + CF)?

- *Pinning a single bond* would let it be cornered; shorts couldn't deliver.
 - *Basket + CF* lets the short pick the cheapest deliverable, while the CF keeps the playing field fair across coupons. Same trick is used in commodity futures with multiple grades or delivery locations.

Determining the futures price (cost of carry, with coupons)

The exact theoretical price is messy (delivery can be any time in the delivery month, any of several bonds). Assume the CTD and delivery date are known. Then it's just the cost-of-carry on a coupon-paying asset:

$$F_0 = (S_0 - I) e^{rT},$$

where S_0 is the cash price of the CTD, I is the PV of coupons paid before delivery, r is the risk-free rate to delivery T .

Determining the futures price — example

CTD = 12% semi-annual coupon bond, $CF = 1.6000$; quoted price = \$115;
 $r = 10\%$ (c.c.); delivery in 270 d. Last coupon 60 d ago; next coupon in 122 d;
subsequent in 305 d. Quoted price?

Hedging with bond futures — the formula

A bond futures contract is (approximately) a forward on the cheapest-to-deliver (CTD) bond. To hedge, **equate dollar durations**:

$$D_P \cdot V_P = N^* \cdot D_F \cdot V_F$$

$$\Rightarrow \boxed{N^* = \frac{D_P V_P}{D_F V_F}}$$

- V_P, D_P : portfolio market value and modified duration.
- D_F : modified duration of the CTD **at the futures' delivery date** — not today. Same logic as the forward: the contract inherits the duration the bond *will have* when it is delivered.
- V_F : futures invoice price per contract = *quoted* price $\times$ contract size.
- Short N^* contracts to hedge a long bond position; long to hedge a short.

Hedge with bond futures — example

Parameter	Value
Portfolio Market Value (V_P)	\$10,000,000
Portfolio Duration in 3 Months (D_P)	6.80 years
Futures Settlement Price (Q_F)	93-02 = 93.0625% of par
Futures Contract Notional Price (V_F)	\$93,062.50
CTD Bond Specification	20-year, 12% p.a. coupon bond
CTD Bond Yield (y_{CTD})	8.80% p.a.
CTD Duration at Futures Expiration (D_{CTD})	9.20 years

① **Portfolio Dollar Duration:** $V_P \cdot D_P = \$10,000,000 \times 6.80 = \$68,000,000$

② **Futures Dollar Duration:** $V_F \cdot D_F = \$93,062.50 \times 9.20 = \$856,175$

③ $N^* = \frac{\$68,000,000}{\$856,175} = 79.423$ contracts

This approach implicitly assumes $V_F \approx \frac{V_{CTD}}{CF_{CTD}}$.

SOFR futures (the modern standard)

- The 3-month SOFR futures contract is the dominant short-term interest-rate future in the US (CME). It replaced the Eurodollar future after LIBOR's retirement at end-2021.
- Cash-settled. **Settlement at the end** of the 3-month period to which the rate applies (in arrears).
- Final settlement = 100 – compounded daily SOFR over the period. With daily SOFR rates r_i applied to d_i days:

$$R_{\text{period}} = \left[\prod_{i=1}^n \left(1 + r_i \cdot \frac{d_i}{360} \right) - 1 \right] \cdot \frac{360}{D}, \quad D = \sum d_i.$$

- One basis-point move in the futures quote = \$25 per contract (\$1M notional, 3-month tenor: $\$1\text{M} \times 0.0001 \times 0.25$).

SOFR futures — example

On Sep 22, 2026, an investor has agreed to pay *3-month SOFR + 200 bp* on \$100M for the 3-month period **Dec 16, 2026** → **Mar 16, 2027**. The Dec 2026 SOFR futures quote today is 96.00 (implied SOFR rate = 4.00%). She sells 100 contracts (\$1M each) to lock in her borrowing rate. Illustrate her effective borrowing rate.

Suppose SOFR compounds to 4.79% over the period. The contract settles at $100 - 4.79 = 95.21$ on **Mar 16, 2027** — the *end* of the reference period, once the daily rates are known.

(Back to) Swaps

Organization of trading

Financial institutions act as market makers, quoting bid and ask fixed rates they stand ready to exchange for floating.

Maturity (years)	Bid	Ask	Swap rate
2	2.97	3.00	2.985
3	3.05	3.08	3.065
4	3.15	3.19	3.170
5	3.26	3.30	3.280
7	3.40	3.44	3.420
10	3.48	3.52	3.500

Percent per annum; payments exchanged quarterly. **Source:** Hull, Table 7.4.

- **Bid:** the fixed rate the market maker *pays* in order to receive floating.
- **Ask:** the fixed rate the market maker *receives* in order to pay floating.
- **Swap rate:** the average of the two — this is what gets published as “the” swap rate.
- The spread is the maker’s compensation, and it *widens with maturity*

Interest rate swap pricing

- For floating rate payer, swap is initially equivalent to going long in a fixed rate bond priced at par, and going short in a floating rate bond priced at par
- For the fixed rate payer, the equivalent cash position is the opposite
- In general, we price a swap by finding the difference between the present value of the fixed and floating rate payments.

Valuing the floating leg: par at every reset

Claim. *Just after each reset, the floating-rate bond is worth N (its notional).*

Setup. Annual reset; continuous compounding; no credit risk.

At reset t the rate R_t is observed — it locks the coupon paid at $t+1$, equal to $N(e^{R_t} - 1)$. Principal N repaid at maturity T .

- *Last period.* At $T-1$, only one payment remains: $N e^{R_{T-1}}$ at T . Discounting back at R_{T-1} (the rate just set for this period):

$$P_{T-1} = N e^{R_{T-1}} \cdot e^{-R_{T-1}} = N.$$

- *One step earlier.* At $T-2$, the bond delivers coupon $N(e^{R_{T-2}} - 1)$ at $T-1$ *plus* a bond worth N at $T-1$. Total = $N e^{R_{T-2}}$, so $P_{T-2} = N e^{R_{T-2}} \cdot e^{-R_{T-2}} = N$.
- By induction, $P_t = N$ at every reset $t = 0, 1, \dots, T-1$. $\square$

The same claim under SOFR

The catch. The induction above needs r_t *known at t* . That is LIBOR. SOFR compounds **in arrears**: the period's rate is not known until $t+1$.

- **Par still holds — by replication.** Compounded SOFR over $[t, t+1]$ is the return on rolling an overnight deposit. Invest N overnight at t and roll: at $t+1$ you hold exactly $N \prod_i (1 + r_i d_i / 360)$, i.e. principal plus coupon. Cost today: N . $\square$
- **Inside the period it differs.** At $s \in (t, t+1)$, $V(s) = N \cdot A(t, s)$ — and $A(t, s)$, the compounding already realised, is a constant at s .

Between resets	LIBOR (set in advance)	SOFR (in arrears)
Next coupon	known at t	unknown until $t+1$
Duration	τ_r (to payment)	≈ 0

Swap pricing: the no-arbitrage condition

- At swap initiation, the present value of the fixed and floating rate payments must be equal.
 - Because entering into a swap is free, and a voluntary exchange has to be fair to both sides.
- Since the present value of the floating payments equals the face value of the floating rate bond, the present value of the fixed rate payments also must equal the face value of the fixed rate bond.
- Thus, the fixed rate on the swap is determined by setting the present value of the future fixed rate payments equal to par.

Swap pricing: implementation

- Suppose you have bootstrapped a zero curve $r_1, r_2, \dots, r_T$ appropriate for discounting the fixed leg (Lec 2).
- The fixed rate c then solves:

$$N = cNe^{-r_1} + cNe^{-2r_2} + \dots + N(1 + c)e^{-Tr_T}$$

- Assume the reset frequency is annual.

$$c = \frac{1 - e^{-Tr_T}}{e^{-r_1} + e^{-2r_2} + \dots + e^{-Tr_T}}$$

- Note:
 - Swaps are priced to be consistent with the zero curve, and hence with implied forward rates, FRAs, and other interest-rate forwards and futures.
 - Over time the swap's value moves with market rates. Like a forward, it is zero sum across the two counterparties.

Swap pricing — example

Q. Zero curve, continuously compounded: 1y = 3.0%, 2y = 3.4%, 3y = 3.7%.
What fixed rate c prices a 3-year annual-pay swap at zero?

A. Discount the three payment dates:

t (years)	1	2	3	Sum
r_t	3.0%	3.4%	3.7%	
$e^{-r_t t}$	0.970446	0.934260	0.894939	2.799645

$$c = \frac{1 - e^{-3r_3}}{\sum_t e^{-r_t t}} = \frac{1 - 0.894939}{2.799645} = \boxed{3.75\%}$$

Example — hedging a bank's balance-sheet risk

Setup. It is December 2026. Southwest savings bank funds \$1M of new *10-year* fixed-rate mortgages (10% p.a.) using rolling *3-month* time deposits (8% p.a. today).

Q. A profit of 2% is locked in for the first 3 months. After that, deposit-rate resets expose the bank to rising rates. Use an interest-rate swap to hedge.

Example (cont'd) — swap vs. futures, and residual basis

Why a swap beats a strip of futures here.

- One contract covers the entire 10 years.
- Long-dated futures (years 5–10) are illiquid; the swap is a single negotiated price.

But the hedge is not perfect. Residual mismatches remain:

- Mortgages *amortize* — principal balance declines over time, while the swap notional is constant.
- Mortgage payments are *monthly*, not semiannual.
- Mortgages can be *prepaid* (call optionality), shrinking the asset side unpredictably.

⇒ Demand for customized swap structures: **amortizing swaps, basis swaps, swaptions**

Specialized swaps I — the amortizing swap

The problem. Southwest's mortgage book *amortizes*. A vanilla swap's notional does not.

Year	1	3	7	10
Mortgage balance	\$1,000M	\$850M	\$600M	\$300M
Vanilla swap notional	\$1,000M	\$1,000M	\$1,000M	\$1,000M
Over-hedge	—	\$150M	\$400M	\$700M

- By year 10 the bank is hedging \$700M of mortgages it no longer own.
- An **amortizing swap** steps the notional down on a schedule matching expected paydown.

Specialized swaps II — the basis swap

The mismatch. KB Kookmin *funds* at 3M KOFR but *lends* to corporates at the 91-day CD rate. Its margin is the **CD – KOFR** spread.

	Receives	Pays	Net
Balance sheet	CD	KOFR	$+(CD - KOFR)$ <i>varies</i>
Basis swap	$KOFR + 18 \text{ bp}$	CD	$-(CD - KOFR) + 18 \text{ bp}$
Total			$+18 \text{ bp, fixed}$

- A **basis swap** exchanges two *floating* rates — there is no fixed leg at all.
- The bank must **pay** the rate it *receives* on its assets and **receive** the rate it *pays* on its funding.

Specialized swaps III — the swaption

The problem. Samsung Electronics plans a \$500M bond issue in 6 months — *if* the board approves the capex. Rates may rise first. But a vanilla swap would have to be unwound if the deal is shelved.

- Buy a $6M \times 5Y$ **payer swaption**: the *right*, not the obligation, to enter a 5-year pay-fixed swap at 4.75%. Cost 80 bp upfront, about \$4M.
- Rates rise to 5.50% and the issue proceeds $\Rightarrow$ exercise, and funding is locked at 4.75%.
- Rates fall, or the board says no $\Rightarrow$ let it expire. The loss is capped at the \$4M premium.

Why does the IRS market exist?

The comparative-advantage story. A 5-year borrowing example:

Borrower	Fixed (5y)	Floating (3M, reset)
AAA Corp	4.0%	SOFR + 0.10%
BBB Corp	5.2%	SOFR + 0.60%
Spread (BBB – AAA)	1.20%	0.50%

- AAA is cheaper in *both* markets — but *comparatively* much cheaper in fixed: 1.20% against 0.50%.
- So each borrows where its edge is largest: AAA fixed at 4.0%, BBB floating at SOFR + 0.60%. Then they swap.
- The gain to share is the difference of the differences:
 $1.20\% - 0.50\% = \boxed{0.70\%}$, split between the two firms and the dealer.

Why isn't that 0.70% arbitrated away?

- The two columns are not comparable commitments. The 5-year fixed rate is **locked for five years**. The floating spread is **reset every three months**, at the lender's discretion.
- If BBB's credit deteriorates, its fixed lender is stuck — but its floating lender simply charges more at the next reset.
- So BBB's *expected* long-run floating cost sits much closer to 5.2% than today's $\text{SOFR} + 0.60\%$ suggests.

What the swap really does

It lets each firm borrow in the market where it has the best *access*, then convert the cash flows into the form it actually wants.

Currency swap — the structure



Two notionals (in two currencies); often *exchanged* at start and end

- A **currency swap** exchanges interest streams in two different currencies
- Unlike an IRS, currency swaps usually *do* exchange principals at start *and* maturity.
- Fixed–fixed currency swap is economically a stack of FX forwards bundled into one contract.
- **Korean angle.** The largest single product in the KRW derivatives market is the **USD/KRW cross-currency basis swap (CCS)** — exchanges KRW vs USD floating cash flows. Korean exporters and KTB-issuing utilities use it daily.

Currency swap example — determining the swap rate K

Setup. A US exporter receives €1M every 6 months for 2.5 years.

$S_0 = 1.2673$ \$/€, $r_{\$} = 5\%$, $r_e = 3\%$ (flat curves in both).

Forward-rate schedule (from $F_t = S_0 e^{(r_{\$}-r_e)t}$):

Maturity (yr)	0.5	1	1.5	2	2.5
F_t (\$/€)	1.2800	1.2929	1.3059	1.3190	1.3323

Swap rate K . Pay €1, receive \$ K each period; choose K so PV of swap = 0:

$$K = \sum_t w_t F_t, \quad w_t = \frac{e^{-r_{\$}t}}{\sum_s e^{-r_{\$}s}}.$$

Equivalently, using $F_t = S_0 e^{(r_{\$}-r_e)t}$:

$$K = S_0 \cdot \frac{\sum_t e^{-r_e t}}{\sum_t e^{-r_{\$}t}} \approx 1.306 \$/e.$$

Swap, or a strip of forwards?

Both perfectly hedge the €1M exposure, both have payoffs known today, and both have the *same present value*. So what actually decides?

Strip of forwards	One swap
Cash flow climbs with the forward curve: \$1.280M → \$1.332M	Constant \$1.306M every period
Five contracts to negotiate, document and unwind	One contract, one negotiation, one credit line
Each date priced off its own forward — transparent	A single quoted rate, with the dealer's spread embedded

- Take the **swap** when predictable cash flow and low administrative cost matter — the usual corporate case.
- Take the **strip** when the underlying exposure is itself uneven, so each date wants matching separately.

Commodity swap — structure and pricing

What it is. Periodic exchange of a fixed price (the swap price) for the floating market price of a commodity. Cash-settled (typical) or physical.

Setup. A company will buy 100,000 barrels of oil in 1 year and again in 2 years. Today's forward prices: $F_1 = \$110$, $F_2 = \$111$. Zero rates: $1y = 6\%$, $2y = 6.5\%$ (c.c.).

Q. What single fixed swap price x per barrel makes the deal fair?

A. PV of the forward strip (per barrel):

$$V = 110 e^{-0.06} + 111 e^{-0.065 \cdot 2} = 103.59 + 97.47 = 201.06.$$

Set x so the constant-payment swap has the same PV:

$$x \cdot e^{-0.06} + x \cdot e^{-0.065 \cdot 2} = 201.06 \Rightarrow x \cdot (0.9418 + 0.8781) = 201.06 \Rightarrow \boxed{x \approx \$110.48}.$$

Commodity swap — what the price reveals

Compare to the forward strip. The swap pays \$110.48 in both years, while forwards would pay \$110 and \$111. So:

- Year 1: the company *overpays* by \$0.48.
- Year 2: the company *underpays* by \$0.52.

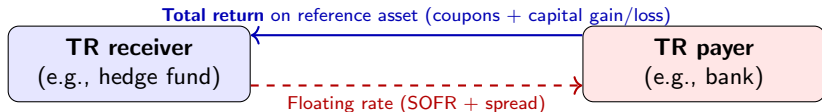
The "overpayment in year 1" is implicitly a **loan from the company to the dealer**. The implied 1-year-forward 1-year-out interest rate:

$$0.483 \cdot e^{r_{1,2}} = 0.517 \Rightarrow r_{1,2} \approx 7.0\%.$$

This matches the implied forward rate from the spot curve:

$$r_{1,2} = (0.065 \cdot 2 - 0.06)/(2 - 1) = 7.0\% \checkmark.$$

Total Return Swap (TRS) — the structure



- One leg: **total return** of a reference asset (bond, loan, equity, index, MBS, EM debt, ...). Total return = coupons + capital gain/loss.
- Other leg: a **floating rate** (typically SOFR + spread).
- No principal exchange. The TR payer need not own the reference asset; if it does, the TRS hedges its cash flows.

TRS — creating leverage (example)

Setup. Three ways to get \$100M of exposure to a reference asset yielding 8.30%.
Floating reference: SOFR = 5.80%; HFs face 100 bp spread to SOFR.

	HF A	HF B	Cash investor
Asset yield	+8.30%	+8.30%	+8.30%
Pay floating (SOFR)	−5.80%	−5.80%	—
Pay spread to SOFR	−1.00%	−1.00%	—
Net asset spread (on \$100M)	+1.50%	+1.50%	+8.30%
Collateral posted	5%	10%	100%
Implied leverage	20:1	10:1	1:1
Interest on collateral	+5.80%	+5.80%	—
Net return on capital	35.80%	20.80%	8.30%

HF A: $[0.083(100) - 0.068(100) + 0.058(5)]/5 = 35.80\%$. The spread over SOFR compensates the bank for the HF's default risk.

Why TRS exists — both sides have a use

TR receiver (long the asset)	TR payer (short the asset)
Off-balance-sheet exposure to an asset class	Hedges an asset it already owns, without selling it
Cheaper to administer — reaches illiquid or restricted underlyings	Short exposure where direct shorting is restricted (e.g. EM debt)
Leverage: \$100M of exposure on \$5M of collateral	Capital relief: less regulatory capital than the cash position

In one line

A TRS slices an asset's *economic exposure* away from its *legal ownership*. Risk-takers buy exposure cheaply with leverage; risk-shedders keep the position on the books while handing off the price risk.

TRS in the wild — Archegos, March 2021

Bill Hwang's family office held roughly **\$100B** of equity exposure — almost all of it through total return swaps, spread across Credit Suisse, Nomura, Morgan Stanley and Goldman.

- He never owned the stock, so the positions appeared in **no ownership disclosure**.
- Each bank saw only its own slice, and judged it well collateralised. None could see the same trade repeated at half a dozen dealers.
- ViacomCBS fell, margin calls went unmet, and the banks sold into a falling market. Credit Suisse lost $\approx$ **\$5.5B**; Nomura $\approx$ \$2.9B.

The point

What the previous slide sells as a virtue — separating *economic exposure* from *legal ownership* — is what made this possible. Leverage was the mechanism; **invisibility** was the cause.

LIBOR and Overnight Rates (Optional)

LIBOR vs. the overnight reference rates

	LIBOR	SOFR / SONIA / KOFR
What it is	Rates <i>estimated</i> by banks, for terms of one day to one year	Overnight rates from <i>actual</i> transactions
Term rate built by	Quoted directly at each tenor	Averaging the overnight series, usually by daily compounding
Rate is known	At the start of the period	Only at the end of the period
Credit risk	Embeds bank credit risk	Treated as risk-free

- The *known at the start* vs *known at the end* row is the one that bites: it is why the floating-leg proof had to be redone, and why SOFR futures settle in arrears.

Life after LIBOR — the legacy-contract problem

USD LIBOR settings ended in **June 2023**. But contracts written years earlier still had time to run: a 20-year swap struck in 2013 matures in 2033.

- You cannot just substitute SOFR. LIBOR embedded **bank credit risk**; SOFR is secured and near risk-free. Swapping one for the other hands value from one counterparty to the other.
- The market's answer was **ISDA fallbacks**: on cessation the floating leg switches to compounded SOFR *plus a fixed credit adjustment spread*, set once from the historical five-year median LIBOR–SOFR gap.
- For 3-month USD that spread was fixed at **26.161 bp** — a permanent constant, so the switch transfers no value on average.
- A few legacy sterling and yen contracts ran on *synthetic* LIBOR, published as term SOFR/SONIA plus the spread, until those were retired too.

Source: ISDA IBOR Fallbacks, spreads fixed 5 March 2021.

Overnight index swaps

- An **overnight index swap (OIS)** exchanges a fixed rate for a floating rate computed from *realised overnight rates*. A modern SOFR swap *is* an OIS.
- **Example.** A two-year OIS initiated 8 March 2022 between Apple and Citigroup:
 - Apple pays **3% per annum** quarterly on \$100M notional;
 - Citigroup pays Apple the **three-month SOFR** reference rate on the same notional.
- Rates quoted with quarterly compounding; ignore day-count and holiday conventions.

OIS cash flows — one possible outcome

Apple pays 3% fixed, receives 3-month SOFR, \$100M notional, eight quarterly exchanges.

Date	SOFR (%)	Floating received (\$000s)	Fixed paid (\$000s)	Net (\$000s)
June 8, 2022	2.20	550	750	−200
Sept. 8, 2022	2.60	650	750	−100
Dec. 8, 2022	2.80	700	750	−50
Mar. 8, 2023	3.10	775	750	+25
June 8, 2023	3.30	825	750	+75
Sept. 8, 2023	3.40	850	750	+100
Dec. 8, 2023	3.60	900	750	+150
Mar. 8, 2024	3.80	950	750	+200

- Each row: floating = $\$100\text{M} \times \text{SOFR} \times 0.25$; fixed = $\$100\text{M} \times 3\% \times 0.25 = \750k .
- **The timing point.** Each SOFR figure is the rate compounded over the quarter *just ended* — known only on the payment date. Under LIBOR, the 2.20% would have been known at inception on 8 March 2022.

Wrap-up — what to take away

- A swap is a **stack of forwards** bundled into one contract — the same no-arbitrage logic, different cash-flow scheduling.
- **The pricing trick.** The floating leg is worth par at every reset, so valuing a swap collapses to pricing the fixed leg alone.
- **The plumbing is not a detail.** Day counts decide what changes hands on each leg; Treasury and SOFR futures supply the forward rates — and hedge a bond book directly.
- **Four flavours: IRS** (fixed for floating), **currency** (two currencies, principals exchanged), **commodity** (fixed for floating price), **TRS** (total return for a floating rate).
- **Big picture.** Forwards (Lec 3–4) and swaps (today) are the *linear* toolkit: payoffs scale one-for-one with the underlying.

Next week: *options* — the first *non-linear* payoff. Right without obligation, asymmetric upside and downside, and the rest of the course.

Reading: Hull, Ch. 9.